23·05·2025 (M)

SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: January 2025 –April 2025 Maximum Marks: 100 Examination: ESE Examination Duration:3 Hrs.		
Programme code: 54 Programme: Honours Programme in Data Science & Analytics	Class: LY-Honours B.Tech	Semester: VII (SVU 2020)
Name of the Constituent College/School K. J. Somaiya School of Engineering	Name of the department: Computer Engineering	
Course Code: 116h54C701	Name of the Course: Advanced Machine Learning	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	20
i)	Given a model with a high bias and low variance and another with low bias and high variance, which model would you prefer for a dataset with limited training data? Explain why.	5
ii)	Compare and contrast the ReLU and sigmoid activation functions in deep learning.	5
iii)	Explain the difference between classification and regression in the context of supervised learning.	5
iv)	Explain the term frequency inverse document frequency (TFIDF) with an example.	5
v)	What is the purpose of a pooling layer in a CNN? Explain the difference between max-pooling and average pooling.	5
vi)	What is a Recurrent Neural Network (RNN), and how does it differ from traditional feedforward neural networks?	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Consider the following simple quadratic cost function for a linear regression model: $J(\theta) = ½ ((\theta^2 - 4\theta + 4))$, Where θ is the parameter to be optimized. a. Calculate the gradient of the cost function $J(\theta)$ with respect to θ. b. Perform one step of gradient descent with the learning rate $\alpha = 0.1$, and an initial $\theta = 2$.	5
ii)	Compare Adam and Stochastic Gradient Descent (SGD) optimizers in terms of their parameter update rule, convergence speed and sensitivity to learning rates.	5
	OR	
Q2 A	Assume you are training several machine learning models on the same dataset for a regression problem. For each model, you calculate the Mean Squared Error (MSE) on the training set and on a validation set. The results are as follows:	10

Model	MSE (Training)	MSE (Validation)
A	5	70
B	20	30
C	7	11

	Answer the following Questions: i. Which model would you say likely under-fits the data? Explain your reasoning. ii. Which model is likely overfitting the data? Explain your reasoning. iii. If you had to choose one of these models to deploy, which one would you choose and why?	
Q 2 B	Solve any **One**	**10**
i)	Consider a problem where you need to classify images of fruits into three categories: apples, oranges, and bananas. Design and draw neural network architecture for multi-class classification of these images. Discuss the pre-processing steps, choice of input features, hidden layers, activation functions, and output neurons for this problem.	10
ii)	Given an input image of size 39x39 with 3 color channels (RGB), and a 3x3 filter with stride 1 and no padding to a CNN Model. Answer the following a) Calculate the output size of the feature map after applying the convolution operation. b) Consider question (a) output as the input to a 2x2 max-pooling layer with stride 2. Calculate the output size after applying the max-pooling operation. c) If the output of the max-pooling layer is passed to a fully connected layer with 20 output neurons, what is the output size after the fully connected layer?	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Explain the purpose of the forget gate, input gate, and output gate in an LSTM. Provide the mathematical equations that define each gate and how they work together to regulate information flow in the network.	10
ii)	Give two examples of real-world tasks where RNNs can be applied. Explain how RNNs process sequential data in these tasks.	10
iii)	Explain the basic working principle of RNNs, focusing on how they maintain the memory of previous inputs.	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Consider a dataset of movie reviews with the sentiment labels positive or negative. i) What features would you extract from the reviews to perform sentiment analysis? ii) Which machine learning algorithms can be used for classifying the sentiment of these reviews?	10
ii)	Give two examples of tasks where Seq2Seq models are used. Explain how the encoder-decoder structure of Seq2Seq helps in these tasks.	10
iii)	How do word embedding techniques capture the semantic meaning of words?	10

Que. No.	Question	Max. Marks
Q5	Write a short note on any **Four**	**20**
i)	Different types of machine learning	5

ii)	Tanh activation function	5
iii)	Convolutional neural network (CNN) vs Artificial Neural Network (ANN)	5
iv)	Phases of NLP	5
v)	Machine learning applications	5
vi)	One hot encoding	5